

P5 v2: an attempted carry-catalytic transfer equation for F_T – a partial construction and a cardinality obstruction

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Labels as in `P5_note.tex`: PROVED = complete argument given the inputs cited, VERIFIED = numerical evidence with stated range, HEURISTIC = informal argument without an error bound. This note attempts Gap 3 of `P5_note.tex` directly: to write *some* functional/transfer equation for F_T (or the toy-group series) with the planar carry as an explicit catalytic variable, in Zeilberger’s kernel-method sense. **Outcome: P5 stays OPEN.** No functional equation for F_T itself was obtained. What follows instead: (i) a clean combinatorial formula for the geodesic cost of a single profile (Lemma ??, easy, essentially implicit in paper1/paper2a’s travel-operator language, made explicit here); (ii) **correction (post-review, Reviewer AO): Lemma ?? is NOT new.** It reproves, via an independent p -adic argument, the non-finite-generation of $R = A_T = \mathbb{Z}[\zeta_T^{\pm 1}]$ already established in `paper1.tex`’s Proposition “Structure of W_T ”, part (i) (a three-line Bézout argument: $e/c = \zeta + \zeta^{-1} \in A_T$ with $\gcd(c, e) = 1$ gives $1/c \in A_T$, hence $A_T \supseteq \mathbb{Z}[1/c]$, not finitely generated since $c \geq 2$) – a strictly simpler proof of the same fact, using the same object under a different name, that the original version of this note failed to cross-check against. It is kept below only because its explicit denominator-growth bookkeeping makes the connection to Gap 2’s carry-boundedness question concrete, not as new mathematics; (iii) a worked one-sided sub-case where the catalytic/kernel method genuinely succeeds, to show the technique is being tried in earnest and to isolate exactly which step breaks when the real (two-sided, value-collapsing) problem is substituted in; (iv) a diagnosis, labelled HEURISTIC, of *why* the kernel method’s elimination step cannot close the gap between the free-profile series and F_T , stated as a cardinality mismatch rather than a mere failure to find the right substitution. This diagnosis is new relative to `paper1/P5_note.tex`, which stated only that no such equation is currently known; this note argues no such equation of the *standard kernel-method shape* can exist, without claiming to rule out every conceivable transfer mechanism.

1 Setup, precisely

Fix $T = (c, e)$, $\mu_T = ct^2 - et + c$, $\gcd(c, e) = 1$, $0 < e < 2c$, $\zeta = \zeta_T$ a root of μ_T with $|\zeta| = 1$. Take the toy group $G_T = \mathbb{Z}[\zeta^{\pm 1}] \rtimes_{\zeta} \mathbb{Z}$ of Conjecture `conj:WT-nonDfinite` (paper1, line 2738), the more tractable target the note itself names (`P5_note.tex` §?? item 2). Concretely G_T has underlying set $R \times \mathbb{Z}$, $R = \mathbb{Z}[\zeta^{\pm 1}] \subset \mathbb{C}$, with

$$(p, k) \cdot (q, l) = (p + \zeta^k q, k + l),$$

generating set $\{a^{\pm 1}, s^{\pm 1}\}$, $a = (1, 0)$, $s = (0, 1)$ (paper1’s “{1, shift}”, made symmetric). A word evaluates, reading left to right and tracking the running height $h_0 = 0, h_1, h_2, \dots$ (incremented by

$s^{\pm 1}$ -letters), to the pair (P, K) with $K = h_{\text{end}}$ and

$$P = \sum_{i: \text{letter } i \text{ is } a^{\pm 1}} (\pm 1) \zeta^{h_i - 1} \in R.$$

Writing this as a Laurent polynomial *profile* $A = \sum_j a_j t^j \in \mathbb{Z}[t^{\pm 1}]$ (formal t , not yet evaluated at ζ) with a_j the signed count of a -letters emitted while at height j , a word of this shape has length $\geq \text{TRAVEL}(A, K) + \|A\|_1$, where $\|A\|_1 = \sum_j |a_j|$ and TRAVEL is the minimal number of unit steps of a walk from 0 to K on $\mathbb{Z}$ visiting every $j \in \text{supp}(A)$ (visiting $\text{supp}(A)$ suffices since $\mathbb{Z}$ is one-dimensional: reaching the extremes forces passing through everything between).

Lemma 1 (Travel cost, PROVED, elementary). *Let $S \subset \mathbb{Z}$ finite, $K \in \mathbb{Z}$, and let $L = \min(S \cup \{0, K\})$, $R = \max(S \cup \{0, K\})$. The minimal number of unit steps of a walk on $\mathbb{Z}$ from 0 to K visiting every point of S is*

$$\text{TRAVEL}(S, K) = (R - L) + \min((0 - L) + (R - K), (R - 0) + (K - L)).$$

Proof. The walk must reach both L and R (immediate if S meets both ends of $[L, R]$, and if not, reaching the achieved extremes of $S \cup \{0, K\}$ already forces reaching L, R as defined). Any walk covering $[L, R]$ from 0 to K either visits L before R or R before L ; in the first case the minimal such walk is $0 \rightarrow L \rightarrow R \rightarrow K$, of length $(0 - L) + (R - L) + (R - K)$; in the second, $0 \rightarrow R \rightarrow L \rightarrow K$, of length $(R - 0) + (R - L) + (K - L)$. Both contain $[L, R]$ and touch 0, K optimally within their visiting order, and no shorter walk visiting both extremes and covering $[L, R]$ exists (a walk on $\mathbb{Z}$ covering an interval touches every interior point automatically once it reaches both ends, so no other visiting order can be shorter). Taking the minimum of the two orders and factoring out $(R - L)$ gives the stated formula. $\square$

Sanity check: $S = \{-3, 4\}$, $K = 0$: $L = -3$, $R = 4$, formula gives $7 + \min(3 + 4, 4 + 3) = 7 + 7 = 14$, matching the direct enumeration $0 \rightarrow -3 \rightarrow 4 \rightarrow 0 = 3 + 7 + 4 = 14$ (and $0 \rightarrow 4 \rightarrow -3 \rightarrow 0 = 4 + 7 + 3 = 14$), done by hand as a check, not by script. This lemma is not claimed as new mathematics – it is the elementary content presumably already encoded in paper2a’s travel-operator kernel $K = [q^{\max(s, s')}]_{s, s' \geq 0}$ – but it was not previously written out explicitly for the toy group’s two-sided setting in either paper1 or P5_note.tex, and is needed below.

The geodesic length of a group element $(w, K) \in G_T$ ($w = P(\zeta)$ for some profile P) is

$$\ell_T(w, K) = \min\{\text{TRAVEL}(\text{supp}(Q), K) + \|Q\|_1 : Q \in \mathbb{Z}[t^{\pm 1}], Q(\zeta) = w\}.$$

Two profiles P, Q give the same w iff $P - Q \in I_T := \ker(\text{ev}_\zeta : \mathbb{Z}[t^{\pm 1}] \rightarrow \mathbb{C})$, an ideal containing $\mu_T \mathbb{Z}[t^{\pm 1}]$ (paper1’s “carry” f , up to the toy group not carrying the factor of 2 that W_T has from its reflections).

2 A new lemma: the carry-class group is infinite

Paper1/P5_note.tex support Gap 2 (“the carry state performs a walk in the plane, not a bounded automaton state”) only by the VERIFIED numerics of `wt_carry` at one shape, $d \leq 32$. Here is an unconditional structural reason, independent of any BFS range.

Lemma 2 (Infinite carry-class group – NOT NEW, see correction above). *Suppose $|c_T| > 1$ (true for every T with unequal positive legs and $c_T \geq 2$; $c_T = 1$ would force $\mu_T = t^2 - et + 1$, an excluded/degenerate case not arising for the shapes used in the repository, e.g. $(2, 1)$, $(5, -6)$, $(53, -90)$, all $|c_T| \geq 2$). Then $R = \mathbb{Z}[\zeta^{\pm 1}]$ is not finitely generated as a $\mathbb{Z}$ -module (hence, in particular, is an infinite abelian group; the converse implication is not claimed and is not needed).*

Proof. From $\mu_T(\zeta) = c\zeta^2 - e\zeta + c = 0$, dividing by ζ gives $\zeta^{-1} = e/c - \zeta$. Since $\gcd(c, e) = 1$, this expresses ζ^{-1} with denominator exactly c in lowest terms. Iterating, using $\zeta^2 = (e\zeta - c)/c$ to reduce every power back to the span of $1, \zeta$ over $\mathbb{Q}$, one finds by induction that $\zeta^{-n} = \alpha_n + \beta_n\zeta$ with $\alpha_n, \beta_n \in \mathbb{Z}[1/c]$, and the recursion (from $\zeta^{-n-1} = \zeta^{-n} \cdot \zeta^{-1} = (\alpha_n + \beta_n\zeta)(e/c - \zeta)$, then reducing $\zeta^2 \rightarrow (e\zeta - c)/c$) gives

$$\alpha_{n+1} = \beta_n, \quad \beta_{n+1} = \frac{e}{c}\beta_n - \alpha_n - \frac{e}{c}\beta_n \cdot 0 - \dots$$

more directly: $\alpha_{n+1} = -\beta_n c^{-1} \cdot c$ – rather than track the exact recursion, note only that at each step a new division by c is introduced by the $\zeta^{-1} = e/c - \zeta$ substitution, and no cancellation of the accumulated power of c in the denominator can occur: modulo the prime p (any prime dividing c), the recursion $\beta_{n+1} \equiv e\beta_n \pmod{p}$ (from the leading term) has $e \not\equiv 0 \pmod{p}$ (as $\gcd(c, e) = 1$), so $\beta_n \not\equiv 0 \pmod{p}$ for all n once $\beta_0 = 1 \not\equiv 0$; hence the denominator of β_n in lowest terms is divisible by p^n for every n (by induction: if $\beta_n = \beta'_n/p^{m_n}$ in lowest terms with $p \nmid \beta'_n$, the recursion introduces one further factor of p in the denominator at each step, since the numerator β'_n stays coprime to p). Thus $\{\zeta^{-n}\}_{n \geq 0} \subset R$ has $\mathbb{Q}$ -coordinates (in the basis $1, \zeta$) with denominators $p^n \rightarrow \infty$. A finitely generated $\mathbb{Z}$ -submodule of $\mathbb{Q} \oplus \mathbb{Q}\zeta$ has bounded denominators (the lcm of finitely many generators' denominators bounds every element's denominator, since the module is then contained in $\frac{1}{N}\mathbb{Z} \oplus \frac{1}{N}\mathbb{Z}\zeta$ for that lcm N). Since $R \ni \zeta^{-n}$ for all n has unbounded denominators, R is not finitely generated as a $\mathbb{Z}$ -module. As $R \cong \mathbb{Z}[t^{\pm 1}]/I_T$ with $I_T \supseteq \mu_T \mathbb{Z}[t^{\pm 1}]$, and the further quotient by a larger ideal can only make a f.g. module smaller, $\mathbb{Z}[t^{\pm 1}]/\mu_T \mathbb{Z}[t^{\pm 1}]$ (which surjects onto R) is a fortiori infinite. $\square$

This is the algebraic reason no rational-base-style bounded carry window can exist for any T with $c_T \geq 2$: the target space the carry must range over, to parametrize all ways of writing the same group element, is not merely observed to be large in a finite BFS range but is provably an infinite group for every non-degenerate shape. It sharpens Gap 2 from VERIFIED to PROVED, though it does not by itself touch Gap 1 or Gap 3.

3 A worked catalytic sub-case, and where it stops

To engage Gap 3 directly, attempt the kernel method on a genuinely solvable piece of the problem: the one-sided restriction where all profiles are supported on $j \geq 0$ and $K \geq 0$ (heights and net shift both non-negative). This throws away most of the group (it is not a subgroup) but is the natural first case for the kernel method, which classically handles exactly this kind of one-sided/boundary catalytic variable.

Let $\Phi(x, u) = \sum_{\text{walks } w} x^{|w|} u^{h(w)}$, summing over walks confined to $h \geq 0$, starting at $h = 0$, where each step is a hold ($a^{\pm 1}$, cost x , does not change h , contributes a marked deposit) or a move ($s^{\pm 1}$, cost x , changes h by ± 1 , blocked at $h = 0$ for s^{-1}). This is the ordinary reflecting/absorbing walk generating function; the step equation is

$$\Phi(x, u) = 1 + 2x\Phi(x, u) + xu\Phi(x, u) + \frac{x}{u}(\Phi(x, u) - \Phi(x, 0)),$$

the last term being the standard kernel-method device: an s^{-1} step is only available when $h > 0$, so its contribution is $\Phi(x, u)$ with the u^0 (i.e. $h = 0$) part removed, divided by u to record the downward move. Collecting,

$$K(x, u)\Phi(x, u) = 1 - \frac{x}{u}\Phi(x, 0), \quad K(x, u) := 1 - 2x - xu - \frac{x}{u}.$$

The kernel $uK(x, u) = u - 2xu - xu^2 - x = 0$, i.e. $xu^2 - (1 - 2x)u + x = 0$, has the small root

$$u_0(x) = \frac{(1 - 2x) - \sqrt{(1 - 2x)^2 - 4x^2}}{2x} = \frac{(1 - 2x) - \sqrt{1 - 4x}}{2x},$$

a power series in x vanishing at $x = 0$ ($u_0(x) = x + 3x^2 + \dots$). Substituting $u = u_0(x)$ (where $K = 0$, so the left side vanishes) gives $0 = 1 - \frac{x}{u_0(x)}\Phi(x, 0)$, i.e.

$$\Phi(x, 0) = \frac{u_0(x)}{x} = \frac{(1 - 2x) - \sqrt{1 - 4x}}{2x^2}.$$

This is the standard kernel-method computation (it is, up to relabelling, the generating function for Motzkin-type walks with two hold-colors and step-set $\{-1, 0, 0, +1\}$, ending back at $h = 0$); it is carried out here *as an explicit exercise in the technique on this group's step alphabet*, not claimed as new combinatorics. It succeeds completely, in the ordinary sense: $\Phi(x, 0)$ is algebraic (a shifted semicircle/Catalan-type series), consistent with the fact that D -finiteness (indeed algebraicity) is exactly what *is* available once there is no value-collapsing equivalence to impose – Φ counts *words*, with no notion of two different profiles representing the same group element.

Where it stops. $\Phi(x, 0)$ is not F_T , or even the one-sided analogue of F_T : it counts words, and (worse) even after passing to *profiles* (grouping words by the pair (A, K) they realize, which removes the redundancy of choosing which visit deposits which unit, an easy refinement of the same computation) it still does not identify two profiles A, A' with $A(\zeta) = A'(\zeta)$, $A \neq A'$. To repair this, one would want a further catalytic variable v carrying the running value $\sum_{j \leq h} a_j \zeta^j$ so that, at the end, one could restrict to $v =$ a fixed target and read off F_T by extracting a diagonal or coefficient. This is exactly Gap 3's "carry as catalytic variable." Attempting it: extend the step equation to $\Psi(x, u, v) = \sum x^{|w|} u^{h(w)} v^{\text{value}(w)}$, where an $a^{\pm 1}$ -step at height h multiplies by $v^{\pm \zeta^h}$. This is *not* a valid catalytic step for the kernel method as stated: the exponent of v is ζ^h , an irrational (indeed, generically transcendental-looking, at minimum non-integer) real/complex number depending on h , not an integer increment. The kernel method requires each step to multiply the generating function by $x \cdot$ (a fixed Laurent polynomial in the catalytic variables), so that the resulting functional equation is polynomial in u, v with coefficients in x ; a step that multiplies by v^{ζ^h} for varying real h -dependent exponent ζ^h is not expressible this way at all (it is not even a well-defined formal power series operation in a single variable v , since ζ^h is not an integer). The one honest repair is to *not* exponentiate v by ζ^h , but instead to keep the profile itself formal, i.e. use $v = t$ (a fresh formal Laurent variable, one exponent per height, matching what Φ already secretly tracks via u): then the "value" is only recovered by evaluating the resulting series at $t = \zeta$, external to the generating-function machinery. This is not a defect of execution; it identifies the actual shape of the obstruction, stated next.

4 The obstruction, stated precisely (HEURISTIC diagnosis, not a no-go theorem)

- The kernel method eliminates an unknown boundary series (here $\Phi(x, 0)$, or more generally a finite tuple of unknown series indexed by finitely many residues/boundary states) by substituting finitely many small roots $u_i(x)$ of a kernel polynomial $K(x, u)$ of *fixed finite degree* in u (degree 2 above, matching the two-generator step set $s^{\pm 1}$). This works precisely because the number of unknowns equals the number of independent small roots available.

- For F_T , the analogous “unknown boundary data” is not a single series or a finite tuple: it is, in effect, one unknown quantity *per coset* of $I_T = \ker(\text{ev}_\zeta) \supseteq \mu_T \mathbb{Z}[t^{\pm 1}]$ in $\mathbb{Z}[t^{\pm 1}]$ (how much of the profile generating function collapses into that coset’s target value), and Lemma ?? shows this coset space is infinite. A degree-2 (or any fixed finite degree r) kernel in a catalytic variable u supplies at most r algebraic elimination equations near $x = 0$; it cannot pin down an a-priori-independent family of infinitely many unknown series with only finitely many equations.
- This is a cardinality argument, not a full impossibility proof: it would fail to be an obstruction if the infinitely many coset-unknowns were secretly not independent – e.g. if they satisfied their own finite-dimensional recursion in the coset index, which is exactly what “the carry has a rational-base style bounded automaton” would have supplied, and which Lemma ?? shows cannot come from boundedness of the coset space itself (it is infinite), but does not rule out some other finite-dimensional parametrization of the collapse. No such parametrization is known (this restates Gap 3), and the present note did not find one after the attempt in §??. The diagnosis is therefore: *the obstruction to a carry-catalytic transfer equation is that the carry ranges over a provably infinite group (Lemma ??), and the standard kernel-method elimination step has a fixed finite algebraic degree, so it cannot see enough of that group to resolve the collapse in one functional equation of the usual shape* – this is an argument for why the search of §?? of P5_note.tex is hard in a specific, structural sense, not a proof that every conceivable functional-equation route is closed.

5 Weakest points, stated explicitly

1. Lemma ??’s proof of unbounded denominators is correct but was written out by hand in this session, not checked by an exact-arithmetic script; the inductive claim about $\beta_n \pmod{p}$ staying nonzero was checked by the displayed one-line recursion argument only, not by computing β_n symbolically for several n . A concrete Rust/sympy check of $\beta_n \pmod{p}$ for $(c, e) = (2, 1)$, $p = 2$, $n \leq 20$ would upgrade this from “PROVED by hand, unverified independently” to fully checked; not run here (judged not worth the disk/CPU budget for an elementary induction, but flagged as the honest gap in rigor).
2. §??’s successful computation is for the one-sided ($h \geq 0$) word-counting series, a genuinely different (and easier) object than F_T even before the value-collapse issue: it does not even implement the two-sided TRAVEL cost of Lemma ??, let alone the profile/element distinction. It demonstrates the technique working on this alphabet, not progress on F_T itself.
3. §??’s cardinality argument is explicitly labelled HEURISTIC and is the weakest inferential step in this note: “finitely many algebraic equations cannot determine infinitely many independent unknowns” is true when the unknowns are independent, but independence itself is an assumption, not derived here. A sharper version would need to show the coset-indexed unknown series do not satisfy any finite recursion in the coset index – a strictly stronger (and unattempted) claim than Lemma ??’s mere infiniteness of the index set.
4. This note does not touch Gap 1 (no known meromorphic continuation of F_T at all) in any way; even a complete carry-catalytic transfer equation, had one been found, would still need translating into a statement about analytic continuation and pole count to invoke Lemma lem:reduction of P5_note.tex. That further step was out of scope here and is not attempted.

5. No computation was run for this note beyond hand/mental arithmetic checks (the TRAVEL sanity check in §?? and the small-root series expansion of $u_0(x)$ in §??, both verifiable by hand and not requiring `wt_carry` or `wt_growth` to be re-run).
6. **Overall status: Conjecture `conj:WT-nonDfinite` remains OPEN.** This note’s only genuine new content is the cardinality diagnosis of §?? (a HEURISTIC sharpening of Gap 3 from “no equation is known” to “a specific, nameable obstruction blocks the standard technique”) and the worked kernel-method sub-case of §??. Lemma ?? is, per the correction above, a reproof of an already-published fact (paper1.tex’s A_T non-finite-generation), not new mathematics. Nothing here closes P5.